

12 (a) Factorise fully $6y^2 - 5y - 4$

$$6y^2 + 3y - 8y - 4$$

$$3y(2y+1) - 4(2y+1)$$

$$(2y+1)(3y-4)$$

$$\underline{y = -\frac{1}{2} \text{ or } y = \frac{4}{3}}$$

$$6y^2 \times 4$$

$$= 24y^2$$

$$\begin{array}{l} / \quad \backslash \\ -8y \quad +3y \end{array}$$

$$y = -\frac{1}{2} \text{ or } y = \frac{4}{3}$$

(2)

(b) Express $\frac{2x+1}{4x} + \frac{7-5x}{3x}$ as a single fraction in its simplest form.

$$\frac{3x \times 2x + 1}{3x \times 4x} + \frac{7 - 5x \times 4x}{3x \times 4x}$$

$$\frac{3x(2x+1) + 4x(7-5x)}{12x^2}$$

$$\frac{6x^2 + 3x + 28x - 20x^2}{12x}$$

$$\frac{-20x^2 + 6x^2 + 3x + 28x}{12x}$$

$$\frac{-14x^2 + 31x}{6x}$$

$$\frac{-7x^2 + 31}{6}$$

$$\frac{-7x^2 + 31}{6}$$

(3)

(Total for Question 12 is 5 marks)